

Shuheng Zhang

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Education

University of California San Diego
Ph.D. Mechanical and Aerospace Engineering

September 2025 - Present
La Jolla, CA, USA

University of Oxford
M.Sc. Mathematical Sciences, Distinction

October 2024 - July 2025
Oxford, UK

Chinese University of Hong Kong, Shenzhen
B.Sc. Mathematics, First Class

September 2020 - July 2024
Shenzhen, China

Publications

Submitted. S. Zhang, M. Diagne, and M. Krstic, “Predictor Feedback for an Age-Structured Population Model with Input Delay,” submitted to the 2026 IEEE Conference on Decision and Control (CDC).

M.Sc. Thesis. S. Zhang, “Random Vortex Method for 2D Viscous Incompressible Flows,” University of Oxford, 2025.

Relevant Projects

Complex-Valued Framework for Nonlinear Opinion Dynamics

2026 - Now

In consultation with Professor Mamadou Diagne

La Jolla, CA, USA

- Proposed a complex Schrödinger-type opinion model separating strength from stance
- Embedded NOD, Kuramoto, and DeGroot as sub-models of one gauge-symmetric system

Neural Tangent Kernel Graphs for Training Curricula

2026 - Now

Independent research project

La Jolla, CA, USA

- Reframed the neural tangent kernel as a weighted graph over training data
- Framed sequential curricula as optimal control of training dynamics on the NTK graph

Hierarchical Feature-Based Synthesis of a Brain Module

2025

Network science research project

Oxford, UK

- Modeled the Drosophila connectome as a directed information-processing network
- Generated scalable synthetic brains from modularity, Hodge-flow, and centrality features

Lifted-Time Stable Inversion in Control Systems

2024

Supervised by Professor Xiaoqiang Ji

Shenzhen, China

- Improved stable-inversion efficiency using a lifted-time formulation

Film Festival Participation

Plant Pears for Your Heirs — Luminar Film Festival

2022, CUHK-Shenzhen

Current — Fisheye Film Festival

2024, UK

Guitar — London International Film Festival (two London festivals)

2024, UK

Awards

Igniting Art Dream Scholarship, Distinguished Alumni Award	2025, CUHK-Shenzhen
St. Peter's College Funding for Artistic Purposes	2025, Oxford
St. Peter's College Travel Grant (documentary <i>Echos</i>)	2025, Oxford
Outstanding Graduating Student of Diligentia College	2024, CUHK-Shenzhen
Best Creativity Award, Documentary Film Competition	2022, CUHK-Shenzhen
Dean's List & Academic Performance Scholarship	2021 & 2022, CUHK-Shenzhen
Bowen Scholarship	2020, CUHK-Shenzhen